

Evolution of Media Discourse on Universal Basic Income in South Korea (2022–2025):

A Mixed-Method Approach Using Topic Modeling and Sentiment Analysis

Project Context

This project was conducted as a voluntary, independent research study to apply and further develop computational and statistical methods in the analysis of media discourse surrounding Universal Basic Income (UBI) in South Korea. It examines how media discourse evolves as UBI-related pilot programs move from political proposals to active implementation. The project is motivated by the need to understand how public narratives in the media evolve as UBI-related policies move from experimental programs to more established forms of policy implementation.

Abstract

In recent years, South Korea has emerged as a valuable empirical context for studying Universal Basic Income (UBI), following the rapid expansion of multiple UBI pilot programs that have generated substantial public attention and media discourse. While existing research has examined the short-term effects of these policies, less attention has been paid to how public narratives and media discourse surrounding UBI have evolved alongside the expansion of pilot programs in South Korea. Accordingly, this study analyzes the evolution of South Korean media discourse on UBI-related cash transfer policies between 2022 and 2025, a period marked by both active UBI policy experimentation and heightened political contestation.

Using a mixed-method approach, the study analyzes 8,930 news articles retrieved from the BigKinds database, combining Latent Dirichlet Allocation (LDA) topic modeling, lexicon-based sentiment analysis, and temporal statistical tests. The results reveal that media discourse is structured around multiple distinct topical frames, with prominent frames centered on electoral contestation and administratively focused policy implementation. Sentiment analysis shows that coverage becomes significantly more negative and contested during presidential election periods, while exhibiting a steady monotonic shift toward more favorable policy-related framing over time.

The findings suggest that media discourse on UBI-related cash transfer policies in South Korea is shaped by a dual dynamic of ongoing political contestation and gradual policy institutionalization. This illustrates how redistribution initiatives can simultaneously function as electoral symbols and as increasingly embedded policy instruments.

Key Findings

Political Volatility: Media sentiment declines significantly during presidential elections, reflecting heightened partisan contestation.

Gradual Institutionalization: Despite short-term volatility, there is a statistically significant monotonic shift toward more favorable, policy-oriented framing over time.

Thematic Shift: Discourse has shifted from short-term, crisis-driven political rhetoric to sustained discussions of concrete regional policy implementation.

Dual Dynamics: UBI-related initiatives function simultaneously as contested electoral symbols and increasingly embedded policy instruments.

Public Repository

All data, code, and analysis for this project are publicly available on GitHub, including cleaned yearly datasets, a reproducible Jupyter Notebook documenting the full analytical workflow, and the web scraping script used for data collection.

<https://github.com/anneminkyung/basic-income-media-discourse-korea>

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Abstract

In recent years, South Korea has emerged as a valuable empirical context for studying Universal Basic Income (UBI), following the rapid expansion of multiple UBI pilot programs that have generated substantial public attention and media discourse. While existing research has examined the short-term effects of these policies, less attention has been paid to how public narratives and media discourse surrounding UBI have evolved alongside the expansion of pilot programs in South Korea. Accordingly, this study analyzes the evolution of South Korean media discourse on UBI-related cash transfer policies between 2022 and 2025, a period marked by both active UBI policy experimentation and heightened political contestation.

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1 Introduction

In recent years, South Korea has implemented a series of Universal Basic Income (UBI) pilot programs and related policies, attracting growing scholarly attention in global debates on UBI (Prochazka 2025; Basic Income Earth Network 2025). Unlike many other countries, South Korea has pursued a variety of quasi-UBI pilot programs and UBI-like experiments that function as policy trials rather than conceptual proposals (Basic Income Earth Network 2025). Representative examples include the Gyeonggi Youth Basic Income (2019), the Gyeonggi Farmers' Basic Income (2021), and the Gyeonggi Artists' Basic Income (2023). These programs operate as categorically targeted cash transfers that are unconditional within defined eligibility groups. Together, these programs have provided South Korea with a valuable empirical context for examining the implementation and societal implications of UBI-like cash transfer schemes, while also contributing empirical evidence to both domestic policy debates and international discussions on UBI (Widerquist 2018; Prochazka 2025). Despite this growing body of empirical evidence, the transition from local experimentation to a national policy framework remains highly contested, as public and media narratives continue to diverge over the long-term legitimacy and implications of UBI (You and Choi 2022; Kang 2025). Accordingly, this study examines how media discourse on UBI in South Korea evolved between 2022 and 2025, a period marked by the shift from political proposal to active policy implementation.

1.1 Problem Statement

As these pilot programs have moved from conceptual proposal to policy implementation in South Korea, they have generated substantial media attention, shaping how such policies are publicly interpreted

and contested. Korean media discourse has frequently articulated concerns commonly associated with global UBI debates, including the perceived utopian nature of the policy, the legitimacy of “income without labor,” concerns over long-term fiscal sustainability, and skepticism regarding the responsible use of transfers by recipients (Widerquist 2018; Standing 2017; Basic Income Earth Network 2025).

However, in South Korea these debates are further intensified by entrenched political polarization. Because Lee Jae-myung, a leading figure of the Democratic Party, has served as the primary advocate and key architect of these pilot programs, media discourse has become highly politicized, with UBI frequently framed through partisan lenses rather than evaluated in terms of policy design or implementation and its effectiveness. Consequently, UBI-related policies are often characterized by labels such as “populist” or “electorally motivated,” alongside persistent concerns regarding fiscal risk. This dynamic leads to media attention that peaks during election cycles while limiting sustained coverage of policy outcomes and implementation mechanics (Park 2025; The Korea Times 2025).

At the same time, the period between 2022 and 2025 presents a critical turning point in South Korea’s policy environment. During this period, UBI shifted from a largely contested political proposal to an expanding set of actively implemented programs across multiple policy domains and local governments. This transition was further solidified following Lee Jae-myung’s electoral victory in the 2025 presidential election and the subsequent introduction of Livelihood Recovery Coupons, which are a quasi-UBI measure that achieved exceptionally high levels of public uptake and satisfaction (Ministry of the Interior and Safety 2025). Such developments highlight a growing dissonance between expanding policy institutionalization and persistently politicized media framing. This raises a critical empirical question: has media discourse evolved to reflect the reality of policy institutionalization, or does it continue to prioritize political contestation even as UBI-related initiatives move toward large-scale implementation?

1.2 Research Question (RQ)

How has media discourse on Universal Basic Income (UBI) and related cash transfer programs in South Korea evolved between 2022 and 2025, and how are these changes reflected in shifts in dominant topics and key policy frames?

2 Methodology

2.1 Policy Context and Scope

The pilot programs in South Korea are quasi-UBI in nature but share key characteristics of Universal Basic Income, including unconditional or broadly targeted cash transfers, limited means-testing, and population-based eligibility (Basic Income Earth Network 2025). As such, these programs have played an important role in shaping public debate and media discourse on income support. Throughout this study, the term “quasi-UBI programs” refers to South Korea’s UBI policy experiments that incorporate core defining features of Universal Basic Income as articulated by the Basic Income Earth Network (2025), without constituting a nationwide or permanent UBI policy. The term “UBI-related cash transfer policies” is used to refer to the broader media discourse that situates these programs within ongoing debates on Universal Basic Income.

Major initiatives analyzed in this study include:

- Gyeonggi Farmers’ Basic Income (경기도 농민 기본소득) (2021)
- Gyeonggi Artists’ Basic Income (경기도 예술인 기본소득) (2023)
- Gyeonggi Youth Basic Income (경기도 청년 기본소득) (2019)
- Disaster Relief Funds (재난지원금) (2020 - 2021)
- Livelihood Recovery Support Funds (민생회복지원금) (2025)

While some of the programs examined in this study were initiated before 2022, the temporal scope of analysis is defined by media discourse rather than policy inception. The 2022–2025 period is selected because it represents a phase in which UBI-related cash transfer programs were most actively debated and framed in Korean news media.

2.2 Data Collection

News articles were collected from BigKinds, the news big data system operated by the Korea Press Foundation (Korea Press Foundation 2025). The dataset covers the period from January 1, 2022, to December 12, 2025. Articles were retrieved when at least one of the following policy-related keywords appeared in the article title:

‘기본소득 (Basic Income)’ OR ‘재난지원금 (Disaster Relief Funds)’ OR ‘민생회복지원금 (Livelihood Recovery Support Funds)’ OR ‘민생회복쿠폰/민생쿠폰 (Livelihood Recovery Coupons)’.

These keywords were intentionally selected to capture both explicit UBI discourse and quasi-UBI pilot programs that function as cash transfer experiments not conditional on labour participation in the South Korean policy context. Although these terms partially overlap conceptually and sometimes co-occur within the same articles, this strategy ensures broad coverage of UBI-related cash transfer discourse while acknowledging the fragmented and evolving nature of related policy debates in South Korea. A title-level filtering approach was adopted to ensure that collected articles were substantively centered on cash transfer policies, rather than mentioning them only within the body text. In Korean news media, policy-related keywords frequently appear in article bodies as background references or comparative examples; restricting retrieval to titles therefore helps reduce topical noise and increases the likelihood that the policy constitutes the primary focus of the article.

Although BigKinds provides metadata and partial textual content, the dataset includes only the first 200 characters of each article’s body. To enable full-text analysis, the complete articles were subsequently collected through web scraping from the original news outlets. During this process, a subset of articles was found to be inaccessible due to deletion, restricted access, or ongoing editorial revisions, and these items were consequently excluded from the analysis. After removing duplicate entries and excluding articles with missing or unavailable full texts, the final analytical corpus comprised 8,930 news articles.

2.3 Analysis Techniques

This study employs a mixed computational approach combining topic modeling, sentiment analysis, and statistical testing.

(1) Topic Modeling

Latent Dirichlet Allocation (LDA; Blei et al., 2003) was applied to identify dominant discourse topics within the corpus. Nouns were extracted using McCab-based morphological analysis (Kudo 2005), and high-frequency functional words were removed. The optimal number of topics was selected based on coherence score comparisons across candidate models.

(2) Lexicon-Based Sentiment Analysis

Media sentiment toward UBI-related cash transfer policies was measured using the KNU Sentiment Lexicon (Park et al. 2018), supplemented with a custom policy-specific sentiment dictionary designed to capture:

- Policy implementation outcomes (e.g., execution, expansion, benefits)
- Perceived economic risks and fiscal concerns (e.g., burden, deficit, inflation)

To minimize framing bias, policy name components that carry inherently positive semantic connotations were excluded from the sentiment analysis.

(3) Temporal and Statistical Analysis

Sentiment scores were calculated at the article level and aggregated on a monthly basis to examine long-term trends in media discourse on UBI-related cash transfer policies.

Both Pearson and Spearman correlation tests were employed. Pearson correlation was used to assess linear trends over time. Spearman correlation was applied to detect monotonic (rank-based) trends, which are more robust to non-linear fluctuations and short-term volatility commonly observed in political media discourse.

In addition, independent-samples t-tests were conducted to compare sentiment distributions between presidential election periods and non-election periods, in order to examine whether political cycles are associated with systematic shifts in media sentiment.

Presidential election periods were defined as follows:

- **2022 election period** : January 1 – June 30, 2022
(Presidential election held on March 9, 2022, in which Lee Jae-myung—widely recognized as South Korea’s most prominent UBI advocate—was not elected.)
- **2025 election period** : December 1, 2024 – June 3, 2025
(Following the impeachment of President Yoon Suk-yeol on April 4, 2025, and the subsequent presidential election on June 3, 2025, in which Lee Jae-myung was elected.)

These election periods were selected not only because they represent moments of heightened political competition, but also because UBI-related cash transfer policies became central campaign issues during these periods. In the Korean context, UBI has been strongly politicized, largely due to Lee Jae-myung’s role as a leading advocate who actively promoted UBI-related policies as his core electoral pledges (Kim, 2022).

Moreover, Lee Jae-myung played a key role in initiating and implementing multiple UBI-related pilot programs, including youth basic income and other local-level cash transfer experiments. As a result, presidential elections involving his candidacy are especially salient moments in which UBI discourse intensifies and becomes more explicitly framed through partisan and ideological lenses.

Thus, focusing on these periods allows the analysis to capture how media framing and sentiment toward UBI-related cash transfer policies shift under conditions of political contestation, as opposed to the more pragmatic and implementation-focused discourse observed during non-election periods.

3 Results

3.1 Topic Modeling Results

To identify dominant frames in Korean media discourse on UBI-related cash transfer policies, Latent Dirichlet Allocation (LDA) was applied to the corpus. Based on coherence score comparison, a six-topic model ($K = 6$) was selected, as it provided the most interpretable and stable topic structure.

Inspection of the highest-probability keywords associated with each topic revealed six distinct and recurrent policy frames. Topics were labeled based on dominant keyword patterns reflecting political contestation, administrative implementation, policy experimentation, crisis response, and regional policy design. Accordingly, the six topics were interpreted as:

- National Politics & Election Discourse (Parties, Candidates, UBI Debate)
- Youth Basic Income Application & Administration (Gyeonggi Province)
- Urban Welfare Policy & Research (Seoul, Social Security, Experiments)
- Local Economic Recovery & Consumption Stimulus (Coupons, Budget, Livelihood)
- Disaster Relief & Recovery Support (Damage, Wildfire, Emergency Aid)
- Rural/Regional Basic Income Pilots (Depopulation, Farming Communities)

3.2 Sentiment Analysis Results

Out of the 8,930 articles, 8,810 contained at least one sentiment-bearing token, while 120 articles were classified as neutral due to the absence of matched sentiment terms.

Table 1 reports the average normalized sentiment scores by dominant topic. Here, a higher mean sentiment score indicates that articles within a topic are, on average, framed using more positively valenced policy-related language, rather than implying normative support or endorsement of the policy itself.

Overall, sentiment was most positive in Urban Welfare Policy & Research and Youth Basic Income Application & Administration, followed by Rural/Regional Basic Income Pilots. These topics are

Table 1: Topic-wise Mean Sentiment and Document Count

Topic Label	Mean Sentiment	Count
Urban Welfare Policy & Research (Seoul, Social Security, Experiments)	0.9442	887
Youth Basic Income Application & Administration (Gyeonggi)	0.8718	913
Rural / Regional Basic Income Pilots (Depopulation, Farming Communities)	0.7926	2826
Local Economic Recovery & Consumption Stimulus (Coupons, Livelihood)	0.7014	1675
Mixed	0.5208	78
National Politics & Election Discourse (Parties, Candidates, UBI Debate)	0.4940	1334
Disaster Relief & Recovery Support (Damage, Wildfire, Emergency Aid)	-0.0120	1217

primarily oriented toward policy implementation, administration, and experimentation. In contrast, National Politics & Election Discourse exhibited comparatively lower sentiment, suggesting more contested and polarized framing. The near-neutral sentiment observed in Disaster Relief & Recovery Support likely stems from its inherent lexical association with negative crisis contexts—such as “damage,” “loss,” and “emergency”—rather than reflecting normative disapproval of the policy itself.

3.3 Temporal and Event-Based Analysis

Long-Term Trends in Media Sentiment

Correlation analysis was conducted to assess long-term trends in media sentiment toward UBI-related cash transfer policies. The Pearson correlation coefficient between time and monthly sentiment was $r = 0.236$ ($p = 0.107$), indicating a weak positive linear association that does not reach conventional levels of statistical significance. This suggests that sentiment does not increase in a strictly linear manner over time.

In contrast, the Spearman rank correlation coefficient was $\rho = 0.301$ ($p = 0.038$) and statistically significant. As Spearman correlation captures monotonic rather than strictly linear relationships, this result indicates that media sentiment has generally increased over time despite short-term fluctuations and non-linear variation.

Taken together, these findings suggest that while media sentiment toward UBI-related cash transfer policies does not follow a smooth linear upward trajectory, it exhibits a statistically meaningful monotonic trend toward more positive framing over the study period.

Presidential Election Period Effects

To examine whether media sentiment differed across political contexts, an independent-sample t-test was conducted comparing sentiment scores during presidential election periods and non-election periods.

The results indicate a statistically significant difference between the two groups ($t = -6.70$, $p < 0.001$), providing strong evidence that media sentiment differs systematically across political periods.

Mean normalized sentiment during presidential election periods ($M = 0.576$) was substantially lower than during non-election periods ($M = 0.673$).

This finding suggests that media discourse on UBI-related cash transfer policies becomes less positively framed during presidential election cycles, reflecting heightened political contestation and polarized framing. In contrast, non-election periods are characterized by relatively more stable and favorable sentiment toward such policies. While these results demonstrate systematic associations between political cycles and media sentiment, they should not be interpreted as direct causal evidence.

Temporal and Topical Visualization

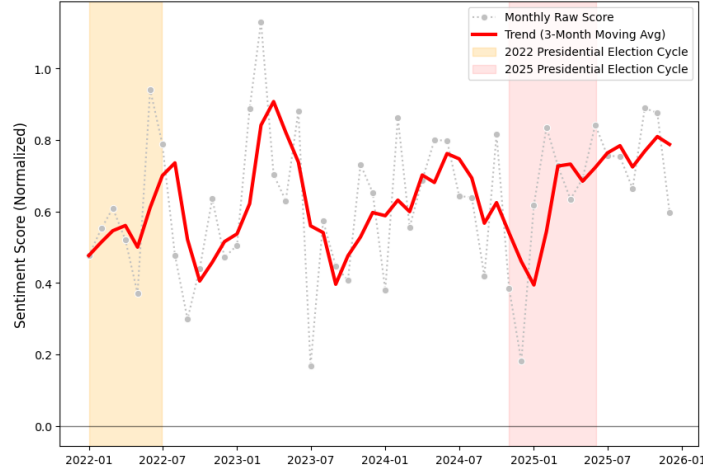


Figure 1: The Evolution of Policy Sentiment: Presidential Election Cycles and Non-Election Periods

Figure 1 illustrates the temporal evolution of media sentiment toward UBI-related cash transfer policies between 2022 and 2025. Monthly sentiment scores (grey dotted line) display substantial short-term volatility, whereas the three-month moving average (solid red line) reveals a gradual upward trend over time. This pattern is consistent with the correlation analysis results, in which the Spearman correlation indicated a significant monotonic increase in sentiment despite the absence of a strong linear trend.

Two presidential election cycles are highlighted in the figure. During these periods, sentiment becomes more volatile and exhibits temporary declines, indicating intensified political contestation and polarized framing of UBI-related cash transfer policies. In contrast, non-election periods are characterized by more stable and consistently positive sentiment levels. Notably, the three-month moving average in Figure 1 captures a distinct recovery following each election cycle, suggesting that the decline in sentiment during electoral periods is temporary, with media discourse returning to more stable levels in subsequent non-election periods.

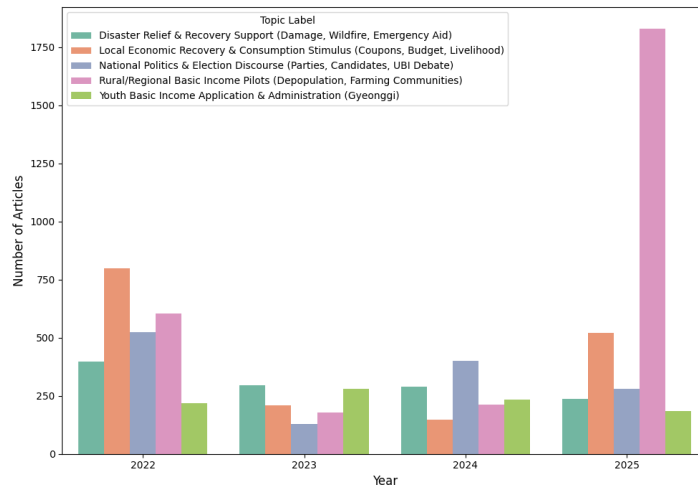


Figure 2: Yearly Shift in Major Topics: Policy Discourse Dynamics

Figure 2 presents the yearly distribution of major discourse topics identified through topic modeling. In 2022, coverage was dominated by local economic recovery and national political discourse, reflecting the legacy of pandemic-era cash transfer policies and electoral debates. In 2023 and 2024, attention shifted toward administratively grounded and regionally focused policy discussions, including youth basic income implementation and rural pilot programs. By 2025, discourse surrounding rural and

regional basic income pilots increased sharply, coinciding with renewed political attention following the presidential impeachment and subsequent election.

Overall, the figure demonstrates a gradual shift from crisis- and election-driven framing toward more sustained discussion of policy experimentation and regional implementation. This shift does not imply the disappearance of political discourse surrounding UBI-related cash transfer policies. Rather, it reflects a rebalancing in which electoral debate increasingly coexists with discussions of policy design, administration, and localized experimentation, suggesting a broadening of the discursive space beyond electoral framing alone. Taken together with the temporal sentiment analysis, this pattern indicates that while political cycles continue to shape media attention, UBI-related cash transfer policies are increasingly discussed as ongoing and operational policy instruments rather than solely as electoral promises.

4 Conclusion

This study shows that media discourse on UBI-related cash transfer policies in South Korea is shaped by a dual dynamic of political contestation and gradual policy institutionalization. Temporal analysis indicates that media sentiment becomes more volatile and less positively framed during presidential election periods, reflecting intensified political competition and polarized framing of redistribution. These findings highlight that UBI and quasi-UBI programs remain closely embedded in electoral politics.

Topic modeling reveals a gradual shift in the substantive focus of media coverage. Beyond national political debate, increasing attention is devoted to administratively grounded and regionally implemented initiatives, such as youth basic income schemes and rural pilot programs. This shift suggests that UBI-related cash transfer policies are not only discussed as campaign promises, but are increasingly treated as concrete policy instruments subject to implementation and evaluation.

In the South Korean context, this coexistence of political contestation and policy experimentation is closely linked to the role of Lee Jae-myung as both a prominent political advocate of UBI and a key actor in local-level policy implementation. Lee occupies a central position in media discourse through which UBI is simultaneously framed as an electoral issue and as an administratively embedded policy experiment. As a result, Korean media discourse reflects both ideological debate and emerging policy normalization, illustrating how UBI-related cash transfer policies can evolve from contested political proposals into sustained areas of policy experimentation. Such experimental approaches align with broader debates on how basic income pilots should be evaluated beyond ideological claims (Widerquist, 2018).

Building on these findings, future work could move beyond media discourse and examine behavioral responses to UBI-like cash transfers using administrative or transactional data (e.g., card spending records). Such an approach would allow future studies to assess whether dominant public narratives—concerns regarding the efficiency of UBI-related spending—are consistent with observed consumption patterns. In addition, analyzing online discourse (e.g., social media posts and news comment sections) could provide further insight into citizen-level framing that may diverge from institutional media narratives.

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